Related Roadmaps

- Computer Science
- Programming Languages
- System Design
- Software Design & Architecture

Data Structures & Algorithms

C#
JavaScript
C++
Java
Python
Go
Rust
Ruby

Pick a Language

Programming Fundamentals

Language Syntax
Control Structures
Pseudo Code
Functions
OOP Basics

What are Data Structures?

Why are Data Structures Important?

Array
Linked Lists
Queues
Stacks
Hash Tables

Basic Data Structures

Common Runtimes

Constant
Logarithmic
Linear
Polynomial
Exponential
Factorial

Time vs Space Complexity
How to Calculate Complexity?
Asymptotic Notation
Big-O Notation
Big-θ Notation
Big-Ω Notation

Algorithmic Complexity

Sorting Algorithms

Bubble Sort
Merge Sort
Insertion Sort
Quick Sort
Selection Sort
Heap Sort

Binary Search
Linear Search

Search Algorithms

Directed Graph
Undirected Graph

Graph Data Structures

Tree Data Structures

Search Algorithms
Breadth First Search
Depth First Search
Shortest Path Algorithms
Dijkstra's Algorithm
Bellman-Ford Algoritm
A* Algorithm
Minimum Spanning Tree
Prim's Algorithm
Kruskal's Algorithm

Binary Trees
Binary Search Trees
AVL Trees
B-Trees
Heap

Tree Traversal
In-Order Traversal
Pre-Order Traversal
Post-Order Traversal

Search Algorithms
Breadth First Search
Depth First Search

Advanced Data Structures

Trie
Segment Trees
Fenwick Trees
Disjoint Set (Union-Find)
Suffix Trees and Arrays

2-3 Trees
B/B+ Trees
Skip List
ISAM

Complex Data Structures

Indexing

Linear
Tree-Based

Brute Force
Backtracking
Greedy Algorithms
Island traversal
Randomised Algorithms
Multi-threaded
Divide and Conquer
Two Heaps
Kth Element
Recursion
Merge Intervals
Dynamic Programming
Cyclic Sort
Two Pointer Technique
Fast and Slow Pointers
Sliding Window Technique

Problem Solving Techniques

Leetcode
Edabit

Platforms to Practice

Continue learning with following roadmaps

Computer Science
System Design